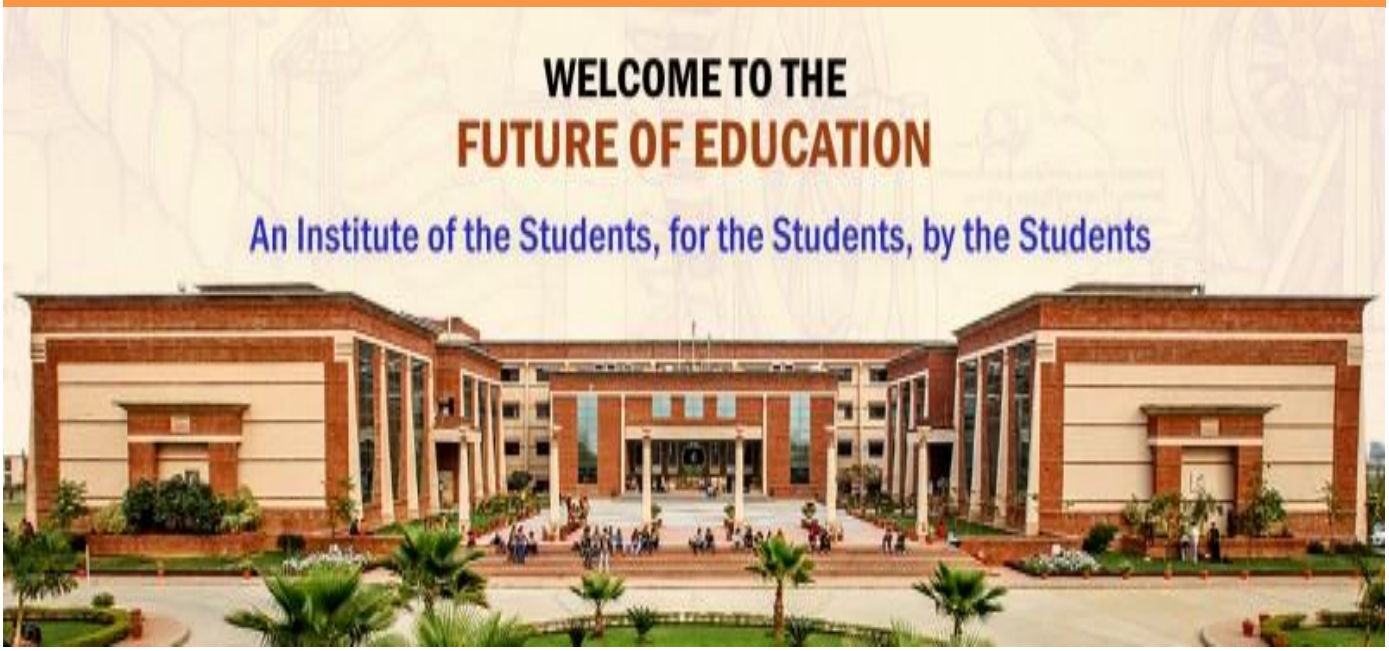
ECHELON INSTITUTE OF TECHNOLOGY

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

NEWSLETTER

JANUARY – JUNE 2026 (ISSUE-2)



VISION AND MISSION OF THE INSTITUTE

VISION

Technical and Management leaders engaged in the evolution of life, being at the frontiers of the continuous technological and administrative breakthroughs, inspired by ongoing exploration of self, society, and nature through self-reflective consciousness by building a culture of inspiration, exploration and growth.

MISSION

M-1 Having a culture of inspiration, exploration, and invention through effective, experiential teaching-learning giving rise to ever-evolving knowledge and wisdom.

M-2 To have self-inspired students ever engaged in continually working upon and sharpening and deepening computational, creative, innovative, and leadership consciousness.

M-3 Having students established in self-reflective consciousness, committed to personal, social & human integrity, and engaged in deep inquiry and conversation, giving rise to shared, inter-subjective human values and consciousness.

VISION AND MISSION STATEMENT OF THE DEPARTMENT

VISION

To be a leading center of excellence in Electronics and Communication Engineering, fostering innovation, research, and leadership through reflective and conscious learning while exploring self, society, and nature.

MISSION

M1: To promote inspiration, innovation, and hands-on learning in Electronics and Communication Engineering for advancing knowledge, technology, and experiential education.

M2: To develop self-motivated students with strong technical, creative, and leadership skills through continuous learning and professional skill enhancement.

M3: To nurture responsible engineers committed to integrity, lifelong learning, reflective practice, and contributing to society through shared human values.

SEMESTER EVENT CHRONOLOGY

A snapshot of major ECE Department activities in the Even Semester 2025–26



Six Days Online

Faculty Development Program

On

Sustainable Energy Technologies for Climate Mitigation

6th to 11th October 2025

ORGANIZED BY

Department of Electronics & Communication Engineering

Echelon Institute of Technology, Faridabad, Haryana, India

ABOUT EIT

Echelon Institute of Technology (EIT), Faridabad, established in 2007, is a premier institution dedicated to academic excellence, innovation, and holistic development. EIT is affiliated to Guru Gobind Singh Indraprastha University, Delhi and is approved by AICTE, Ministry of Education, and Government of India. At EIT, we strive to nurture technical and managerial leaders who drive change and innovation through a culture of inspiration, exploration, and continuous learning. We place a strong emphasis on experiential and outcome-based education, providing students with the practical skills and knowledge necessary to thrive in an ever-evolving global landscape.

ABOUT THE DEPARTMENT

This department was established during the inception of the institute in 2007 as the Department of Electronics and Communication Engineering (ECE). The Department was set up to nourish the students into highly skilled engineers with extensive knowledge in the field of Electronics & Communication. The department is well equipped with all the gadgets, instruments and software required for experimental work. Project work throughout the course strengthens the necessity for a combination of practical and theoretical skills and abilities. The students are also encouraged to participate in extracurricular activities and clubs.

ABOUT THE FDP

The Online Faculty Development Programme (FDP) on "Sustainable Energy Technologies for Climate Mitigation" is designed to empower faculty members and researchers with advanced knowledge of sustainable and renewable energy systems. This programme aims to provide in-depth understanding of emerging technologies such as solar, wind, biomass, hydrogen, and energy storage systems, highlighting their critical role in combating climate change and achieving global sustainability goals. It seeks to enhance educators' ability to integrate climate mitigation and clean energy topics into academic curricula and research, while promoting interdisciplinary learning and fostering collaboration among academia, industry, and policy institutions. Participants will develop a research-oriented mindset focused on innovative, low-carbon technologies and practical solutions for energy sustainability. Through this FDP, faculty members will gain comprehensive insights into clean energy systems, enabling them to effectively teach and mentor students in related subjects and projects. Additionally, attendees will be equipped to contribute to institutional initiatives aligned with national and global energy policies, promoting collaborative research, publications, and proposal writing in the renewable energy sector. Ultimately, the programme aims to cultivate change agents who drive sustainability awareness and action within their institutions and communities. Expected outcomes include improved teaching practices, enhanced guidance for student projects in sustainable energy, and strengthened academic-industry partnerships to support India's climate action goals and its transition to a greener future.

ABOUT ATAL ACADEMY

The AICTE Training and Learning (ATAL) Academy was established with the vision to empower faculty members in achieving the goals of higher education, including access, equity, and quality. The Academy conducts a series of workshops in key areas identified by AICTE, aiming to enhance the technical education landscape across the country. Its objectives include setting up a dedicated platform to plan and facilitate the delivery of quality technical education, supporting technical institutions in promoting research, innovation, and entrepreneurship through targeted training programs, and emphasizing the empowerment of technical teachers and technicians by leveraging information and communication technologies. The Academy also utilizes platforms like SWAYAM and other resources to deliver effective training. Furthermore, it provides diverse opportunities for professional development and knowledge exchange through workshops, orientation programs, learning communities, peer mentoring, and various faculty development initiatives. Additionally, the ATAL Academy supports policymakers in integrating relevant training components to meet evolving educational needs.

ORGANIZERS

CHIEF PATRON

Mr. Prabhat Agrawal

(Honorable Chairman ,EIT, Faridabad)

PATRON

Dr. Arvind Kumar Pandey

Director, EIT Faridabad

COORDINATOR

Dr. Anurag Dwivedi

Dean, R&D, EIT, Faridabad

CO-COORDINATOR

Dr. Pramod Singh

(Dean First Year, EIT, Faridabad)

09–14 FEBRUARY 2026

ATAL Faculty Development Programme

“Sustainable Energy Technologies for Climate Mitigation”

A six-day online ATAL FDP was organized by the Department of ECE. The programme brought together faculty and experts around emerging sustainable energy technologies, climate mitigation, renewable energy and research opportunities. The FDP strengthened faculty capacity for contemporary teaching and research.

ECHELON INSTITUTE OF TECHNOLOGY



16 FEBRUARY 2026

From Design to Reality: 3D Printing Workshop

Organised by IEEE Student Branch, EIT

A hands-on workshop introduced students to the journey from digital design to physical 3D-printed objects. The session was conducted for B.Tech 4th Semester and BCA 4th Semester students at F-35 from 10:35 AM to 12:15 PM, with emphasis on additive manufacturing and practical exposure.



MARCH 2026

HackIndia 2026

The Professional Development Cell organized HackIndia 2026, providing students an opportunity to participate in an innovation-driven Web3 and Artificial Intelligence hackathon environment. The activity encouraged teamwork, problem solving, technology application and solution development.

Seminar on Mitigation of Air Pollution in NCR at Institution of Engineers New Delhi



MARCH 2026

ECE students, along with faculty member Mr. Kumail Hasan Naqavi, attended the seminar on "Mitigating Air Pollution in NCR: Causes, Impacts on Vulnerable Population and Solutions," gaining valuable insights into air-quality challenges, their impacts, and sustainable mitigation strategies.

VISUAL HIGHLIGHTS

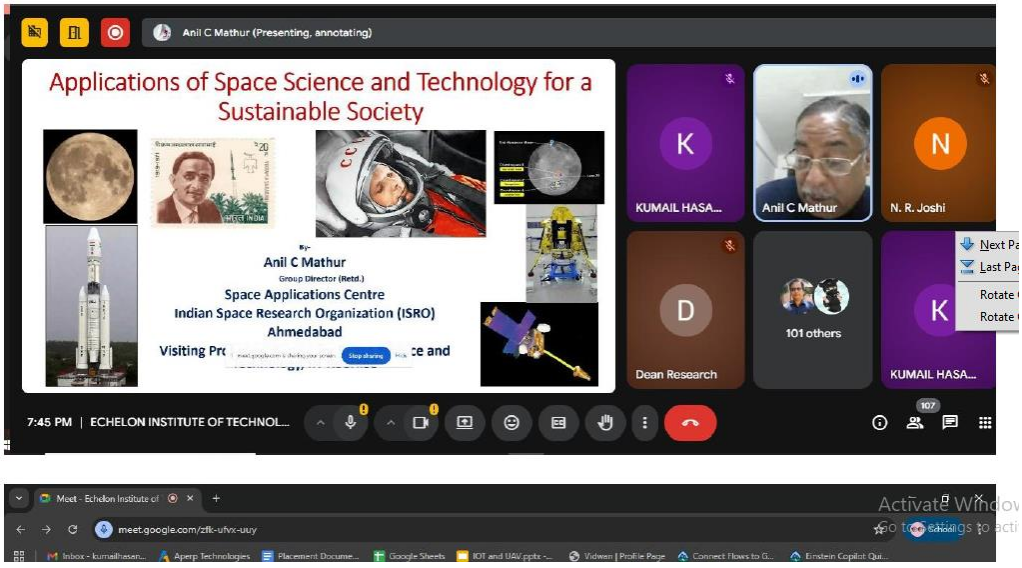
Hands-on learning, expert interaction and student engagement



3D Printing Workshop — expert-led practical interaction



Students engaged in the 3D printing learning session



Expert interaction on Applications of Space Science and Technology for a Sustainable Society

MPACT & WAY FORWARD

Strengthening ECE through practical learning, innovation and professional exposure

Student-Centric Learning — Activities were designed to complement curriculum delivery with demonstrations, practical exposure, expert interaction and collaborative learning.

Emerging Technology Exposure — 3D printing, sustainable energy technologies, space applications, Web3 and Artificial Intelligence provided students and faculty with exposure to rapidly evolving technology domains.

Professional & Research Orientation — Faculty development and expert-led sessions encouraged continuous professional learning, interdisciplinary research and application-oriented thinking.

Innovation Culture — Hackathon and hands-on activities created opportunities for students to convert ideas into prototypes, solutions and collaborative projects.

Department Perspective

The Even Semester 2025–26 reflects the department's continued emphasis on experiential learning and professional development. By integrating academic instruction with workshops, faculty development, expert sessions and innovation platforms, the ECE Department aims to prepare students for technology-driven careers and meaningful contributions to society.

LOOKING AHEAD

More workshops • stronger industry interaction • research & innovation • student projects • emerging technology exposure

INTERNATIONAL CONFERENCE HIGHLIGHT

ICASTSD VIRTUAL-2026 • 29–30 MAY 2026

International Conference on AI, Smart Technologies and Sustainable Development

(ICASTSD VIRTUAL-2026)



Organized by Echelon Institute of Technology (EIT), Faridabad, in association with Inspira Research Association (IRA), Jaipur.

CONFERENCE AT A GLANCE

The International Conference on AI, Smart Technologies and Sustainable Development (ICASTSD VIRTUAL-2026) was organized by Echelon Institute of Technology on 29–30 May 2026. The conference brought together academic, research and technology perspectives around Artificial Intelligence, smart technologies and sustainable development, strengthening EIT's research and innovation ecosystem.

Key Focus Areas: Artificial Intelligence • Smart Technologies • Sustainable Development • Research & Innovation • Interdisciplinary Collaboration

PLACEMENT & CAREER DEVELOPMENT HIGHLIGHT

TECHNICO INDUSTRIES LTD. • CAMPUS PLACEMENT DRIVE • 30 MAY 2026



Selected Students with Dean R&D & HoD ECE Dr. Pramod Singh, Training &Placemen Head Dr. Sangeeta

Company	**Technico Industries Ltd.**
Role	**Technical Positions**
Date	**30 May 2026**
Time	**10:00 AM**
Venue	**Seminar Hall, II Floor, Academic Block**
Eligibility	**B.Tech / ME / ECE / EEE**

1. Introduction

Echelon Institute of Technology, Faridabad, organized a campus placement drive by Technico Industries Ltd. on 30 May 2026. The drive provided eligible students with an opportunity to interact with an industry recruiter and explore technical career opportunities.

2. Objective of the Drive

To provide students with direct exposure to industry recruitment processes; to facilitate employment opportunities in technical roles; to enable students to demonstrate their academic knowledge, technical skills and professional communication; and to strengthen industry–academia interaction and career readiness.

3. Student Participation and Selection

Students from eligible B.Tech/ME/ECE/EEE programmes participated in the placement process. The drive involved interaction with the recruiting organization and assessment for suitable technical positions. As an outcome of the drive, four ECE students successfully secured placement opportunities with Technico Industries Ltd.

4. Achievement and Impact

The successful placement of four ECE students is a significant achievement for the Department of Electronics & Communication Engineering. It reflects the students' technical preparedness and professional competence and demonstrates the value of continuous industry-oriented training and career development activities. The drive also contributed to strengthening the institute's industry interface and creating pathways for future recruitment opportunities.

5. Conclusion

The Technico Industries Ltd. campus placement drive was a valuable industry–academia engagement. The successful selection of Tuneer, Vineet, Neeraj and Fardeen is a matter of pride for Echelon Institute of Technology and the ECE Department. The department extends its congratulations to the selected students and wishes them success in their professional careers.

PLACEMENT SUCCESS

4 ECE STUDENTS SUCCESSFULLY PLACED

Tuneer • Vineet • Neeraj • Fardeen

STARTUP & INNOVATION HIGHLIGHT

CIRCUITCRATE • SONIPAT STARTUP SUMMIT 4.0 • 12–13 FEBRUARY 2026

CircuitCrate Team at Sonipat Startup Summit 4.0

An inspiring exposure to entrepreneurship, innovation and startup collaboration



CircuitCrate, an ECE startup initiative, represented Echelon Institute of Technology at Sonipat Startup Summit 4.0.

INNOVATION IN ACTION

The CircuitCrate team, an ECE startup from Echelon Institute of Technology, participated in Sonipat Startup Summit 4.0 on 12–13 February 2026. The event provided valuable exposure to the startup ecosystem and enabled the team to interact with entrepreneurs, innovators, industry representatives and fellow startup teams.

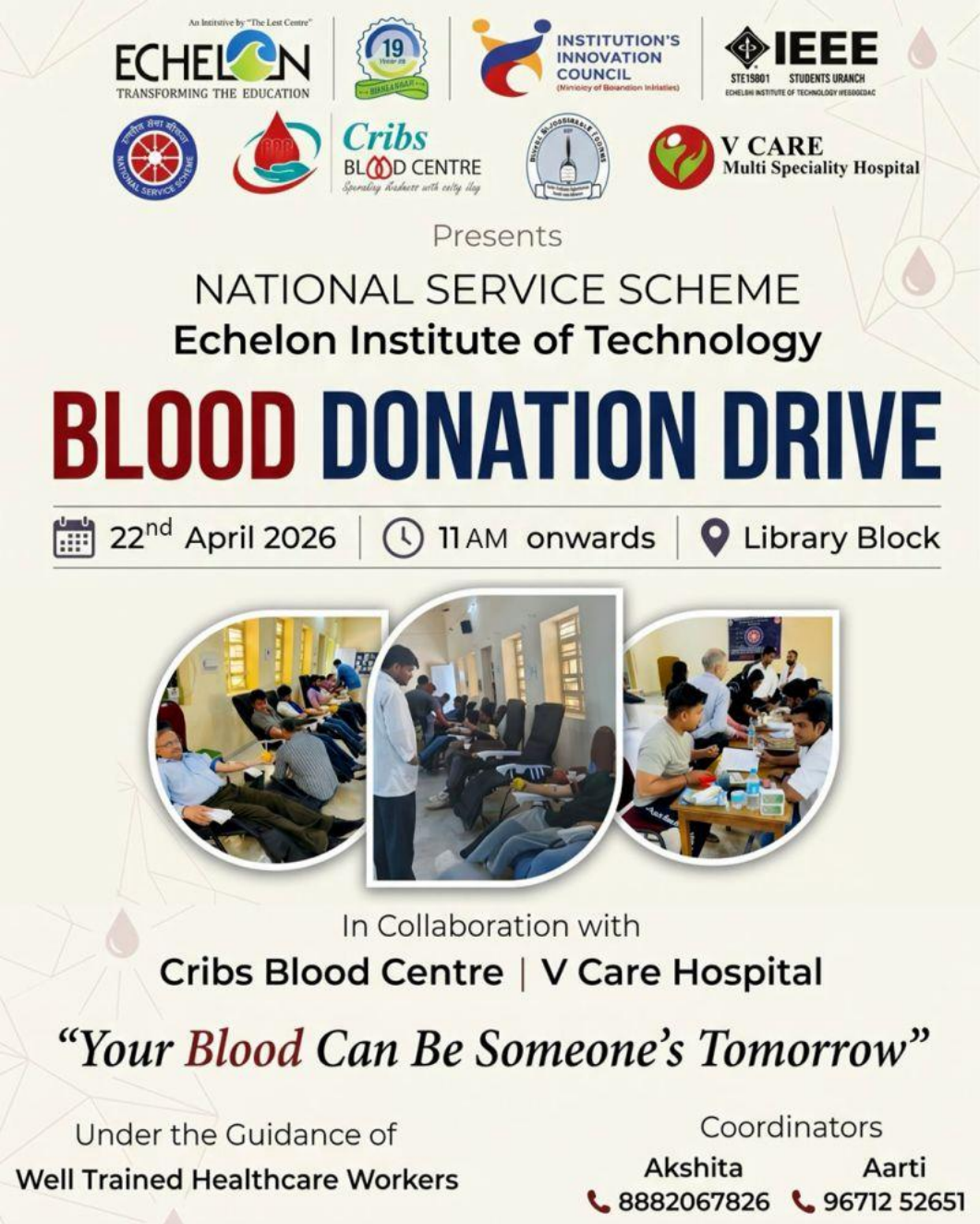
The team gained valuable insights from experienced entrepreneurs, explored opportunities for collaboration and networking, and strengthened its motivation to scale CircuitCrate as an innovative technology and education venture.

Key Takeaways

- Exposure to the startup and innovation ecosystem.
- Interaction and networking with entrepreneurs and industry stakeholders.
- Insights into building and scaling impactful technology solutions.
- Opportunities for meaningful collaboration and future partnerships.
- Greater motivation to develop CircuitCrate as an ECE-led innovation venture.

12–13 FEB	CIRCUITCRATE	STARTUP SUMMIT
2026	**ECE Startup**	**Sonipat 4.0**

Blood Donation Drive

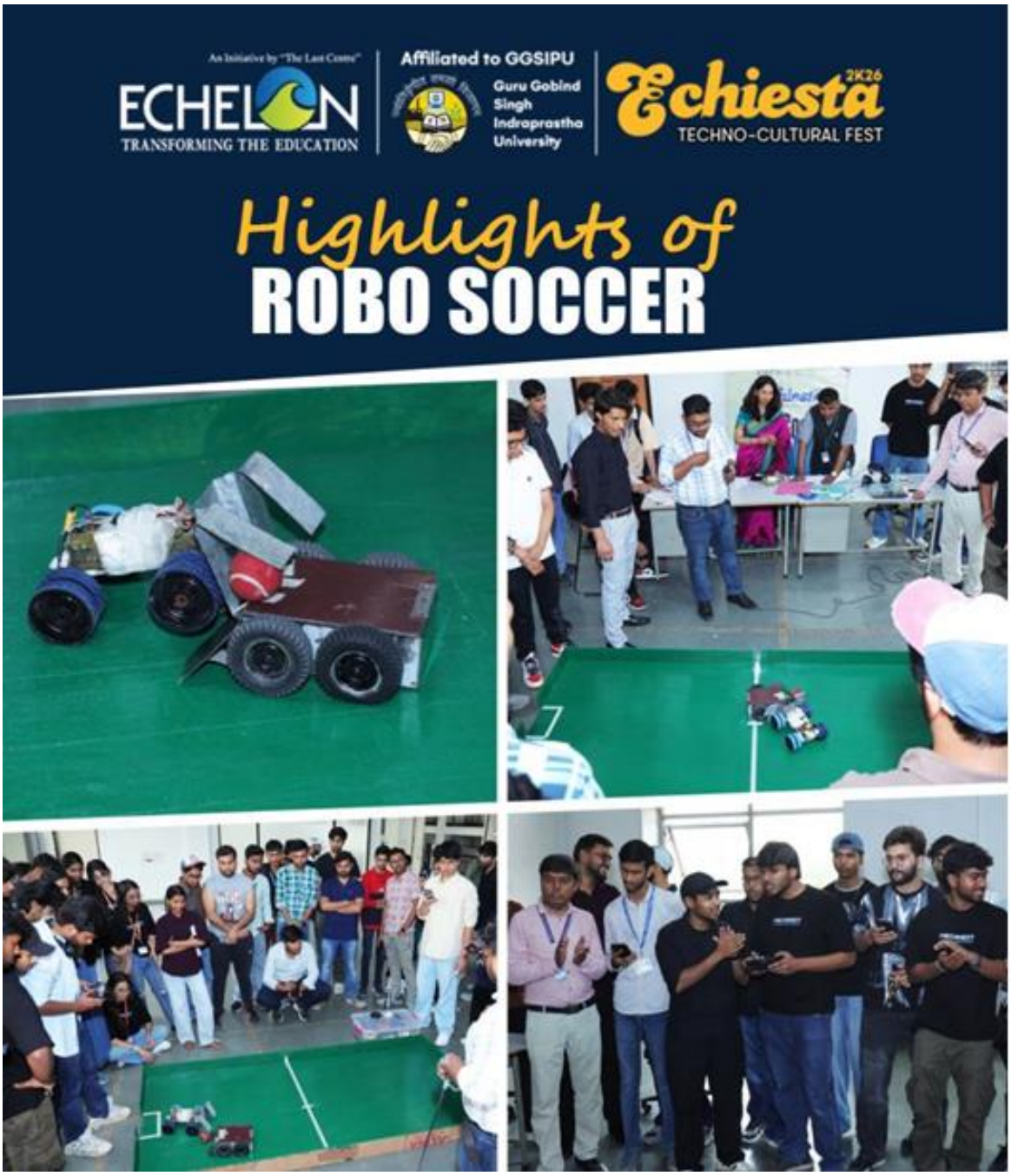


Echelon Institute of Technology organized a Blood Donation Drive in collaboration with Cribs Blood Centre and V Care Hospital. The initiative promoted social responsibility and community service, with students and staff contributing to a meaningful healthcare cause under the guidance of trained healthcare workers.

ECHIESTA TECHNO-CULTURAL FEST

Robo Soccer – Echiesta 2K26

Robo Soccer showcased the practical application of robotics, embedded systems, electronics, control and teamwork. Students demonstrated and operated custom-built robotic vehicles in a competitive arena, providing an engaging platform for hands-on engineering, problem-solving and innovation.



TECHELON 2026



TECHELON 2026 – A Grand Celebration of Experiential Education

The event witnessed active participation from students of B.Tech, BBA, and BCA programs who explored the importance of experiential education and industry-oriented learning in today's rapidly evolving world.

The event was honored by the presence of distinguished guest speakers:

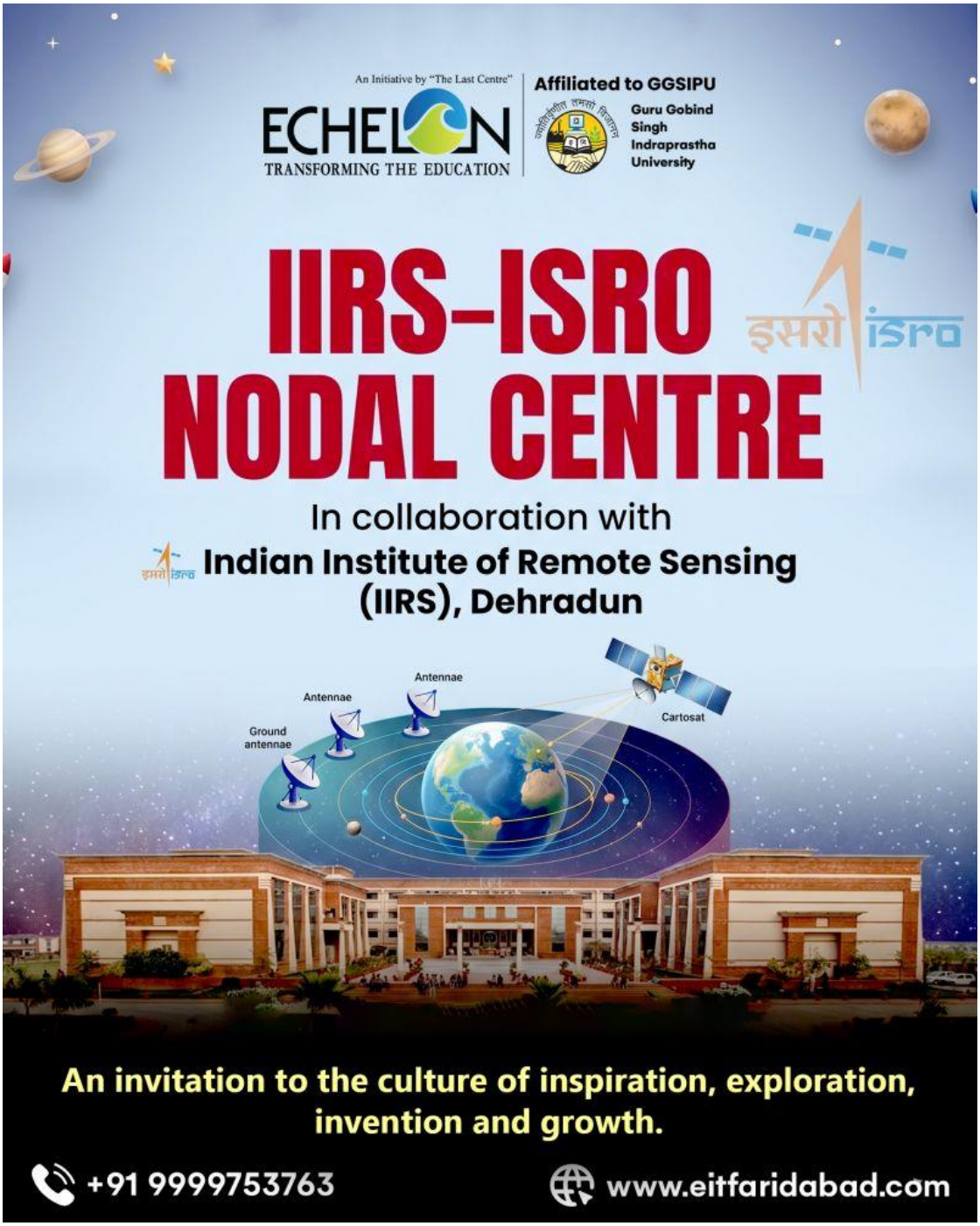
- Mohammad Rihan

Director General, National Institute of Solar Energy

- Prof. Asheesh K Singh

Professor, MNNIT & IEEE Senior Member

EIT–IIRS–ISRO Nodal Centre Initiative



Echelon Institute of Technology strengthened its academic and technological outreach through the IIRS–ISRONodal Centre initiative in collaboration with the Indian Institute of Remote Sensing (IIRS), Dehradun. The initiative creates opportunities for students and faculty to engage with remote sensing, geospatial technologies and space-technology applications through learning and capacity-building activities.

ECE TECHNOLOGY WATCH

NEW DEVELOPMENTS IN ELECTRONICS & COMMUNICATION • TECHNICAL ARTICLES

Connecting classroom concepts with the technologies shaping the next generation of engineering

6G / IMT-2030	AI + Communication, ISAC, ultra-low latency, ubiquitous connectivity
CHIPLETS	UCIe 3.0, heterogeneous integration, high-bandwidth die-to-die links
OPTICAL INTERCONNECTS	Silicon photonics, optical I/O, AI data-center networking
NTN / SATELLITE	5G-NTN, regenerative payloads, satellite-IoT and terrestrial integration
EDGE AI	Local inference, intelligent sensors, low-power accelerators and distributed intelligence
SMART SENSING	Integrated sensing + communication for positioning, mapping and environment awareness

WIRELESS COMMUNICATION

6G: When the Network Becomes a Sensor and an AI Platform

The next generation of mobile communication is being shaped around IMT-2030 (6G). The emphasis is moving beyond simply increasing data rate toward networks that combine communication, sensing, computing and artificial intelligence. In February 2026, ITU-R WP 5D completed draft minimum technical performance requirements for IMT-2030, covering six usage scenarios including AI and communication and integrated sensing and communication (ISAC).

Why it matters for ECE: ECE engineers will need to combine RF/microwave design, antennas, signal processing, machine learning and network architecture. ISAC is particularly important because the same radio infrastructure can support both data transmission and functions such as positioning, localization, object detection and mapping.

Key concepts: AI-native networking • ISAC • high-frequency radio • beamforming • distributed intelligence • energy efficiency

SEMICONDUCTOR & VLSI

Chiplets and UCIe 3.0: The Rise of Modular Electronics

Large monolithic chips are increasingly being complemented by chiplet-based system-in-package architectures. UCIe 3.0 provides an open die-to-die interconnect standard with 48 GT/s and 64 GT/s data-rate options, along with improvements in manageability, power efficiency and system flexibility. This approach allows different functional dies to be integrated into a single package.

Why it matters for ECE: Chiplet design creates opportunities for ECE students in digital VLSI, high-speed PCB design, signal integrity, power integrity, packaging and hardware architecture. It also demonstrates why electrical design continues to matter even as semiconductor systems become highly heterogeneous.

Key concepts: Chiplets • SiP • die-to-die links • signal integrity • power delivery • advanced packaging • UCIe

ADDITIONAL DEPARTMENT & INSTITUTE HIGHLIGHTS

Academic • Innovation • Outreach • Technical & Co-curricular Engagement

TECHNICAL ARTICLES

SHORT READS FOR ECE STUDENTS & FACULTY

PHOTONICS & HIGH-SPEED COMMUNICATION

Optical Interconnects: Moving Data with Light

As AI and high-performance computing demand rapid movement of data between processors, memory and network devices, electrical interconnects face increasing bandwidth, reach and power challenges. Silicon photonics and co-packaged optical I/O are emerging as important approaches. In February 2026, ITU released the ION-2030 framework describing how optical networks can evolve for AI, 6G, data centres, broadband and integrated sensing and communication.

Why it matters for ECE: This development connects electromagnetics, optical communication, semiconductor devices and high-speed electronics. Future ECE engineers can work across lasers, photodetectors, waveguides, optical transceivers, signal integrity and network architecture.

Key concepts: Silicon photonics • optical transceivers • photodetectors • waveguides • co-packaged optics • fibre communication

SATELLITE & WIRELESS COMMUNICATION

5G Non-Terrestrial Networks: Extending Connectivity Beyond the Tower

Non-terrestrial networks (NTN) integrate satellites or high-altitude platforms into communication systems. 3GPP's continuing NTN evolution includes improved terminal performance, uplink capacity, service-area notification and regenerative payload concepts. NTN can support coverage in remote areas and complement terrestrial networks for IoT, mobility and resilient communications.

Why it matters for ECE: ECE students can explore satellite links, antenna systems, Doppler compensation, propagation modelling, link budgets, RF front ends and network integration. NTN is a strong interdisciplinary area connecting communication engineering with aerospace and embedded systems.

Key concepts: LEO satellites • link budget • Doppler • RF front end • NTN • satellite IoT • regenerative payload

EMBEDDED SYSTEMS & IOT

Edge AI: Bringing Intelligence Closer to the Sensor

Edge AI moves selected AI inference tasks from remote cloud servers toward sensors, embedded processors and local gateways. Processing data near the source can reduce latency, communication load and dependence on continuous cloud connectivity. This is especially relevant for smart cameras, industrial IoT, robotics, drones and intelligent monitoring systems.

Why it matters for ECE: For ECE, Edge AI creates a natural convergence of embedded systems, microcontrollers, sensors, DSP, communication and machine learning. Students can build practical projects in which a sensor captures data, an embedded processor performs inference and only useful information is transmitted.

Key concepts: TinyML • NPUs • sensor fusion • embedded ML • low-power processing • IoT • real-time inference

SUGGESTED STUDENT MINI-PROJECTS

- Design an ISAC-inspired radar/communication simulation using MATLAB or Python.
- Compare a conventional monolithic SoC architecture with a chiplet-based architecture.
- Build an Edge-AI sensor node using ESP32/Raspberry Pi and a lightweight ML model.
- Calculate a LEO satellite communication link budget including free-space path loss and Doppler shift.
- Model an optical-fibre link and study attenuation, dispersion and receiver performance.